

MEDIATEK

EINT IDH Share

Outline

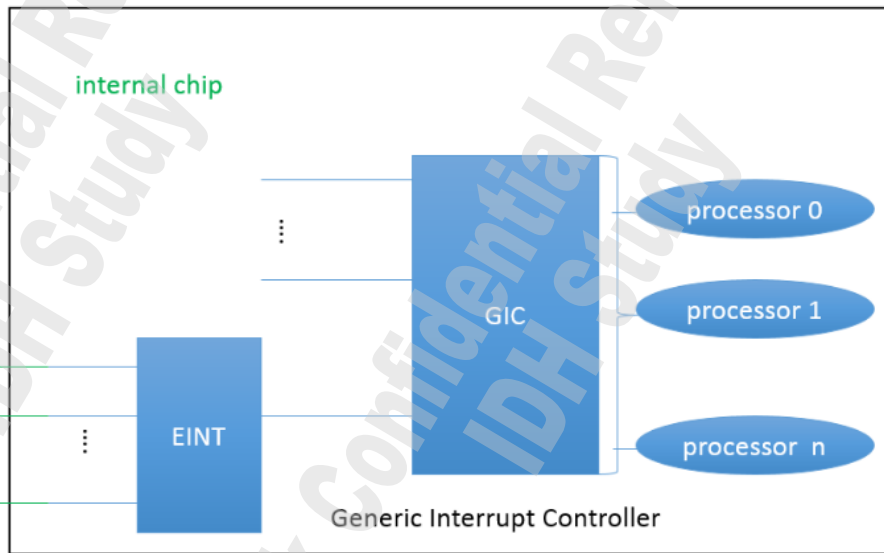
- ❑ Eint Introduction
- ❑ HW Design
- ❑ CIRQ/Deint
- ❑ SW Flow
- ❑ Virtio eint flow
- ❑ Eint Rregister Introduction
- ❑ Debug tips

Eint Introduction

- ❑ Eint: External Interrupt
- ❑ Chained Interrupt controller with GIC
- ❑ Irqdomain, virq
- ❑ Driver depend on gpio driver, as function of gpio
gpio register platform driver, eint not platform driver

HW Design

Chip Design Block



Chip Design Block

GPIO电平变化



EINT硬件检测到变化, 设置EINT status reg pending bit



EINT发信号给GIC, GIC设置自己的pending bit



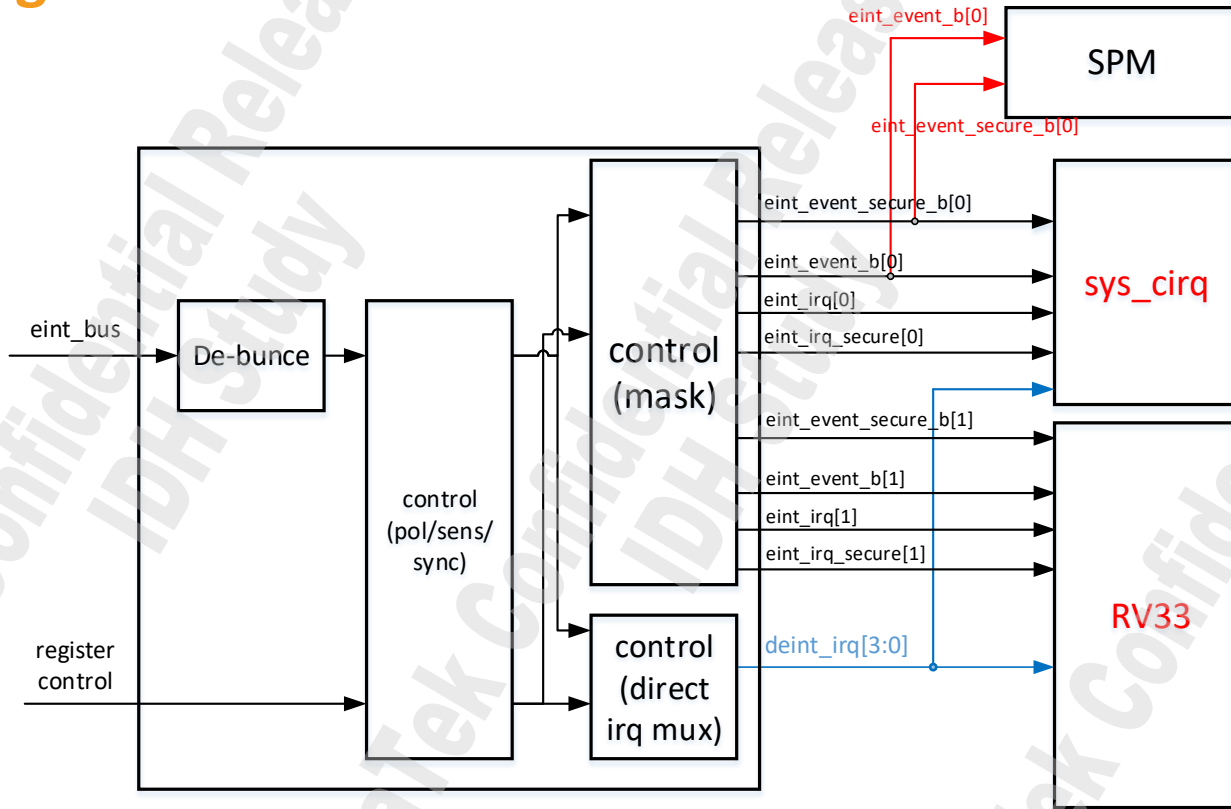
CPU响应GIC中断, 进入EINT中断处理函数



EINT handler:

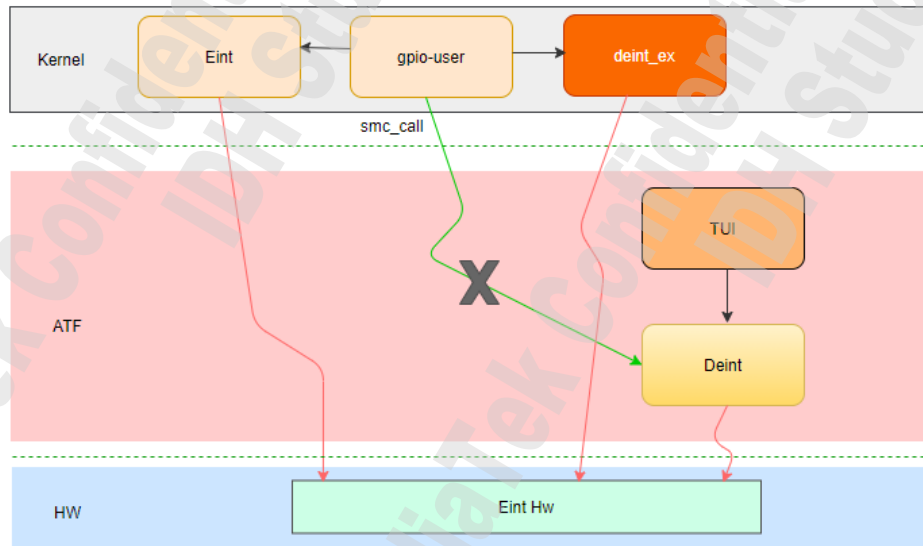
- mask & ack GIC pending bit
- 读取EINT status reg, 遍历pending bit
- mask & ack EINT pending bit
- 处理具体GPIO/EINT事件

EINT Diagram Block



Deint

- 1: 4 gic spi
- 2: design in ATF for secure user(smc_id can not used for NSEL1 user)
- 3: can add kernel driver to use deint but not advised



CIRQ (circular IRQ)

常见的Low power scenarios有MCDI → SODI → Dpidle → Suspend.

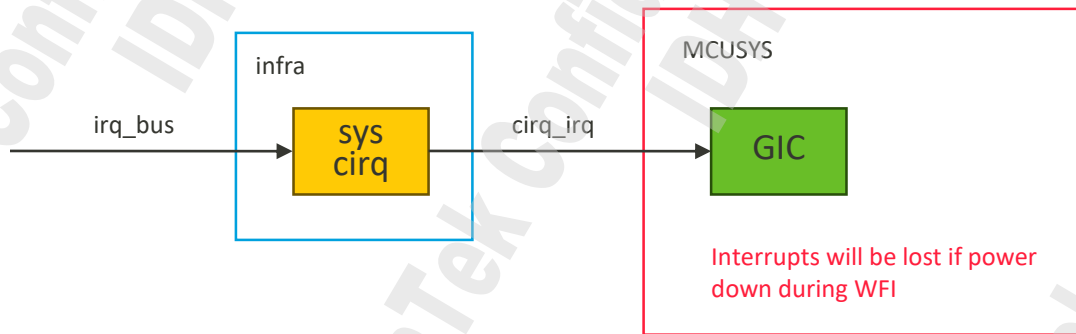
除了MCDI不会关MCUSYS外 (但会enable HW DCM, 有可能导致GIC freq 随CPU降低到13M), 其余都是会关MCUSYS; 而GIC在MCUSYS里, 会随之掉电。

这样单靠GIC就会丢掉恰好处于low power状态的edge中断、如keypad, 这就需要处于infrasys domain的CIRQ帮忙保存中断, 并在GIC随CPU醒来的时候, flush 保存的中断to GIC. 让GIC处理。

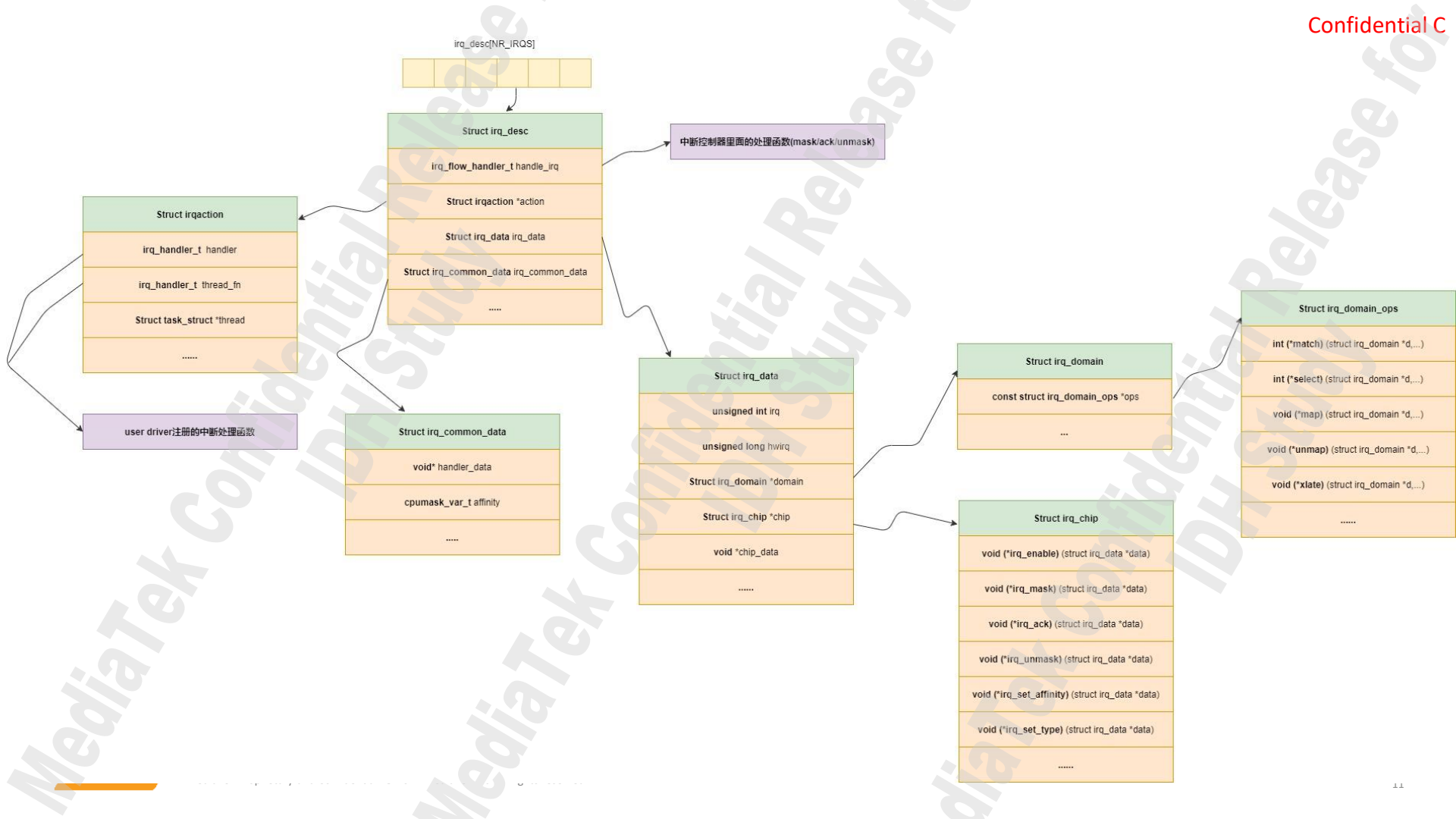
If w/o cirq, 当中断触发时:

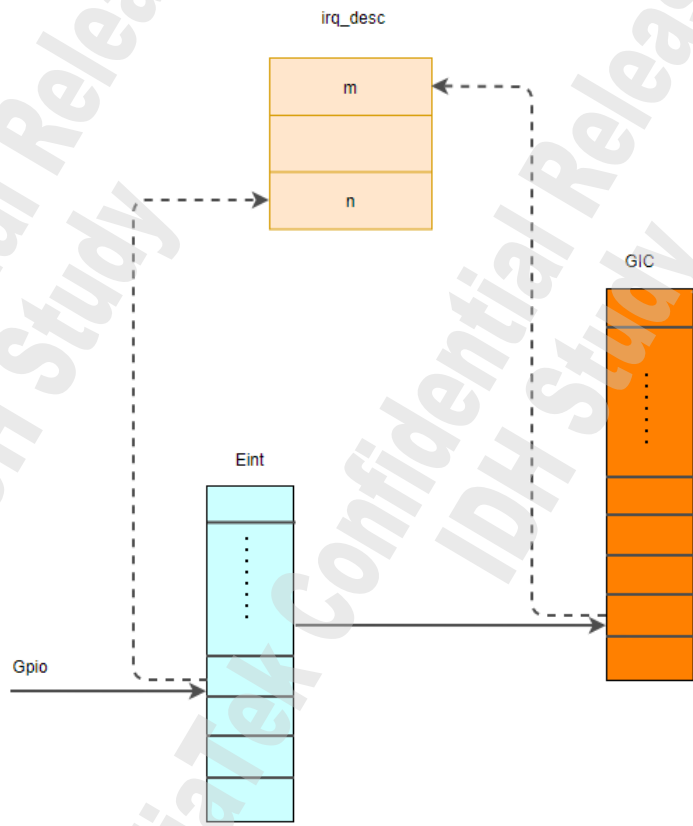
如果这个中断不是SPM的wakeup src, 那就只能等CPU醒后再处理, 这样如果是edge的就会丢失(level信号不会丢失, 是因为CPU不醒来处理的话, 其hw状态一直在); 如果这个中断是SPM的wakeup src, 如kp edge interrupt, 会先通过SPM将CPU唤醒,

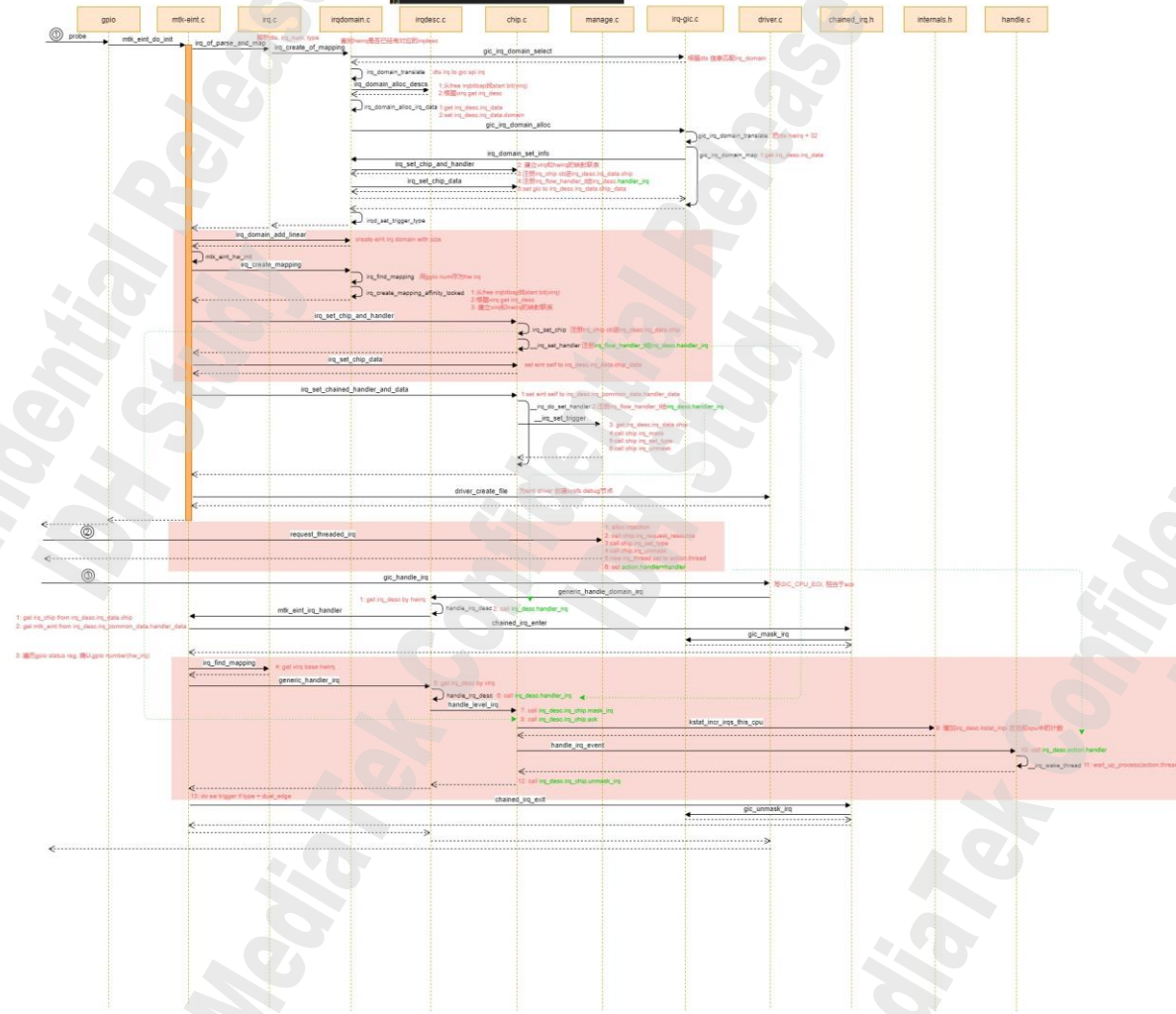
但是醒来后由于edge信号已经丢失, 就会不加以处理, 相当于白白醒来一次, 而没有做音量+/-的处理, 导致按键不灵敏的bug.



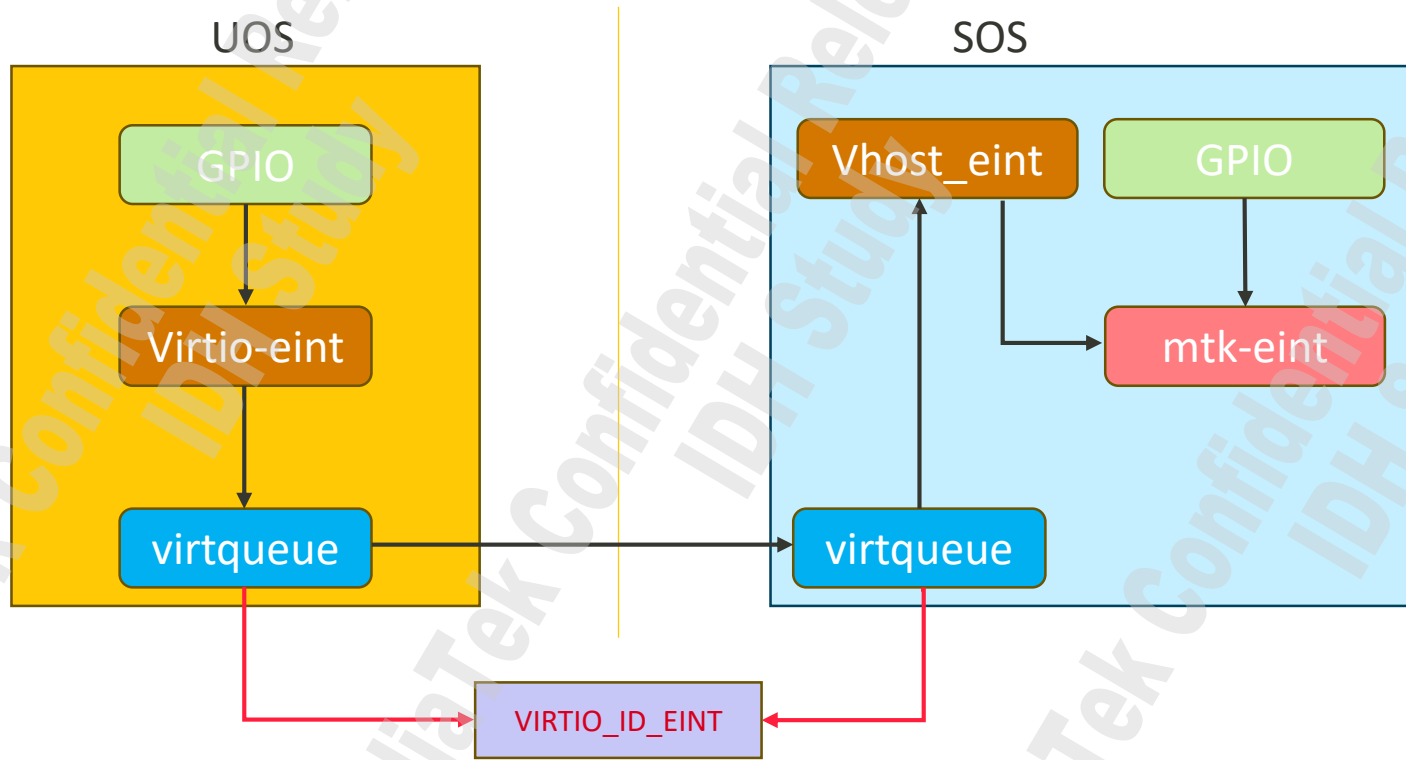
SW Flow



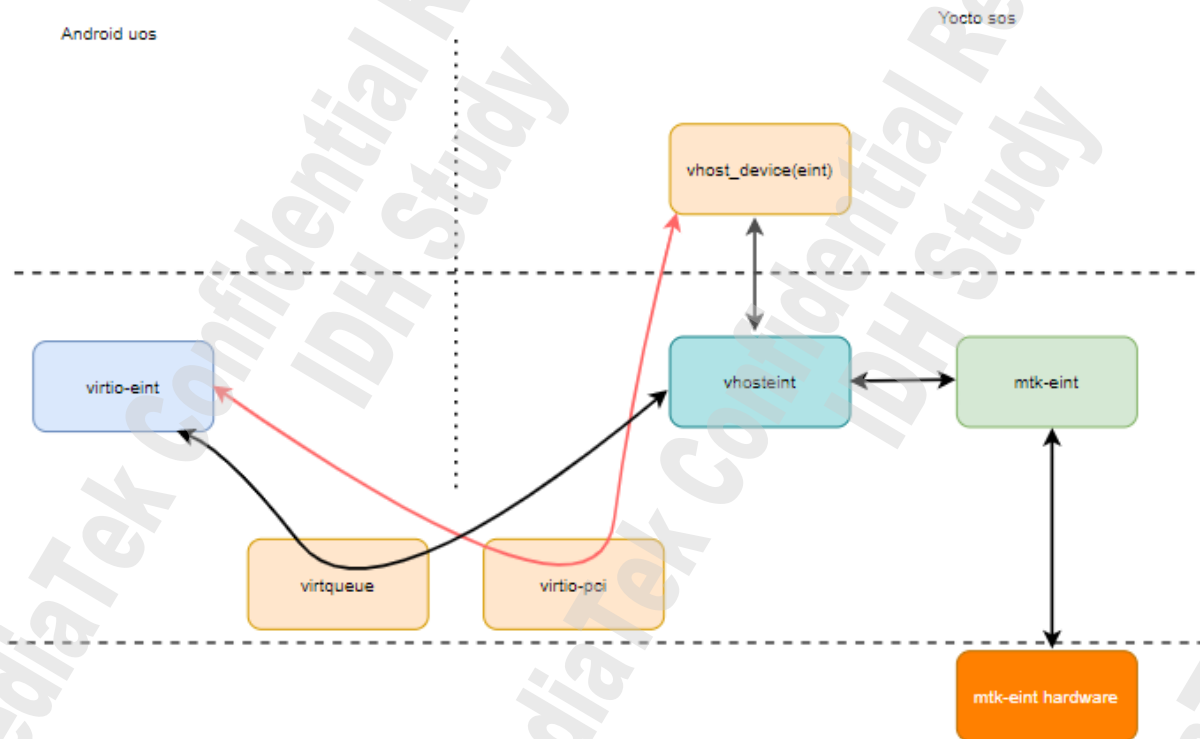




Virtual Eint(vhost)



vhostEint

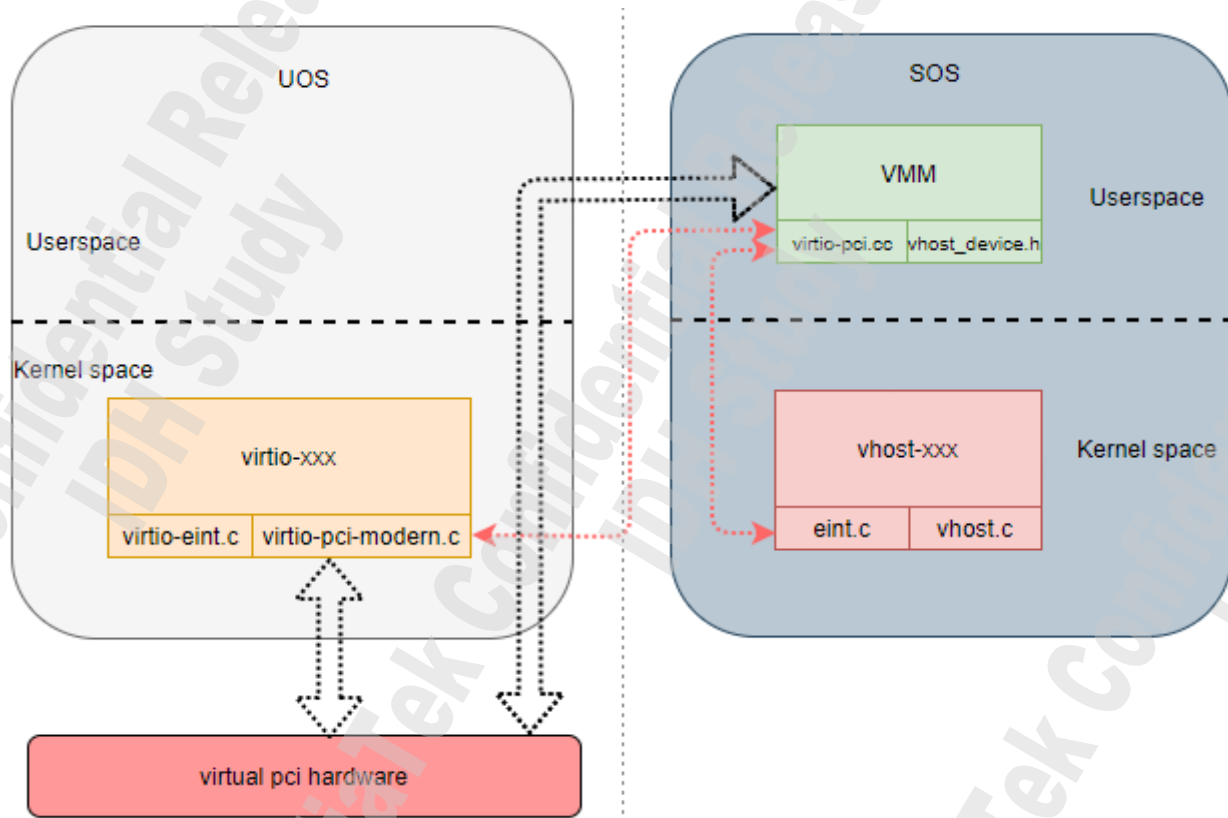


1: VMM(vhost_device)在每个virtqueue notify BAR地址上创建一个I/O event触发点, create eventfd, 把eventfd传给vhost

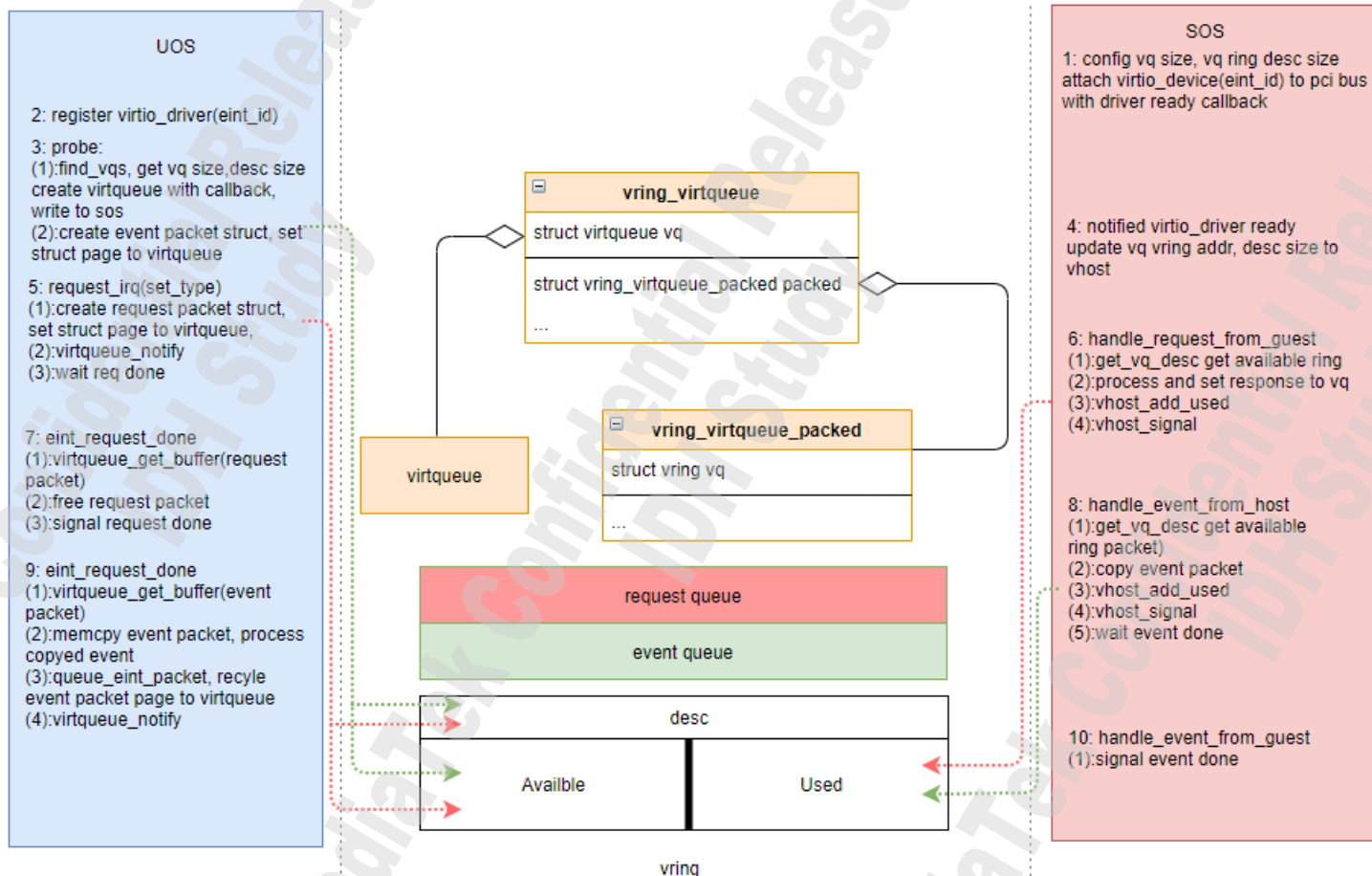
2: guest对virtqueue的notify, 本质上就是写BAR 某个Offset

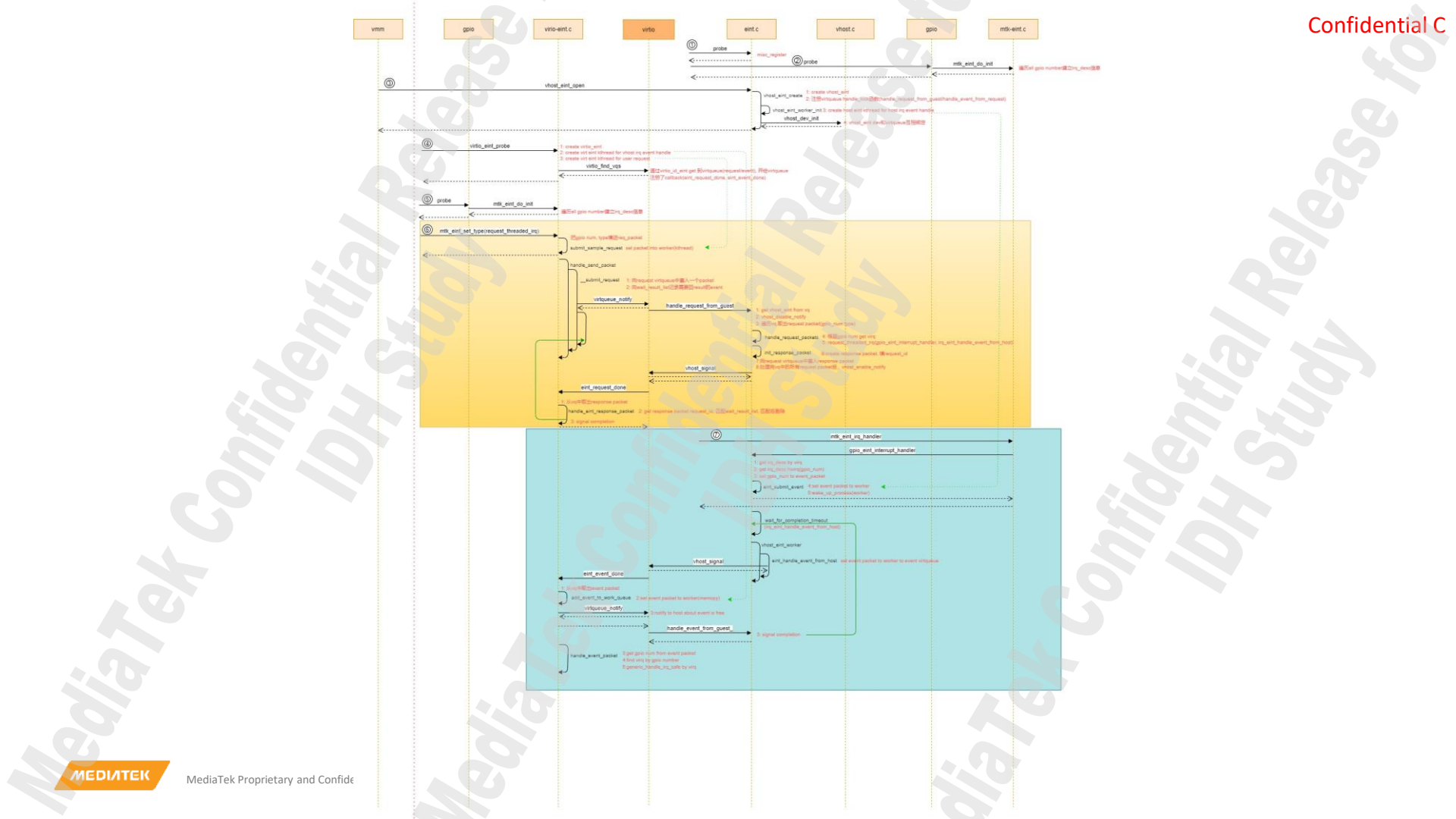
3: 之后guest每写一次notify BAR, 就会产生对应event fd的event, vhost poll监听到之后, 回调对应的handle_kick

Virtqueue

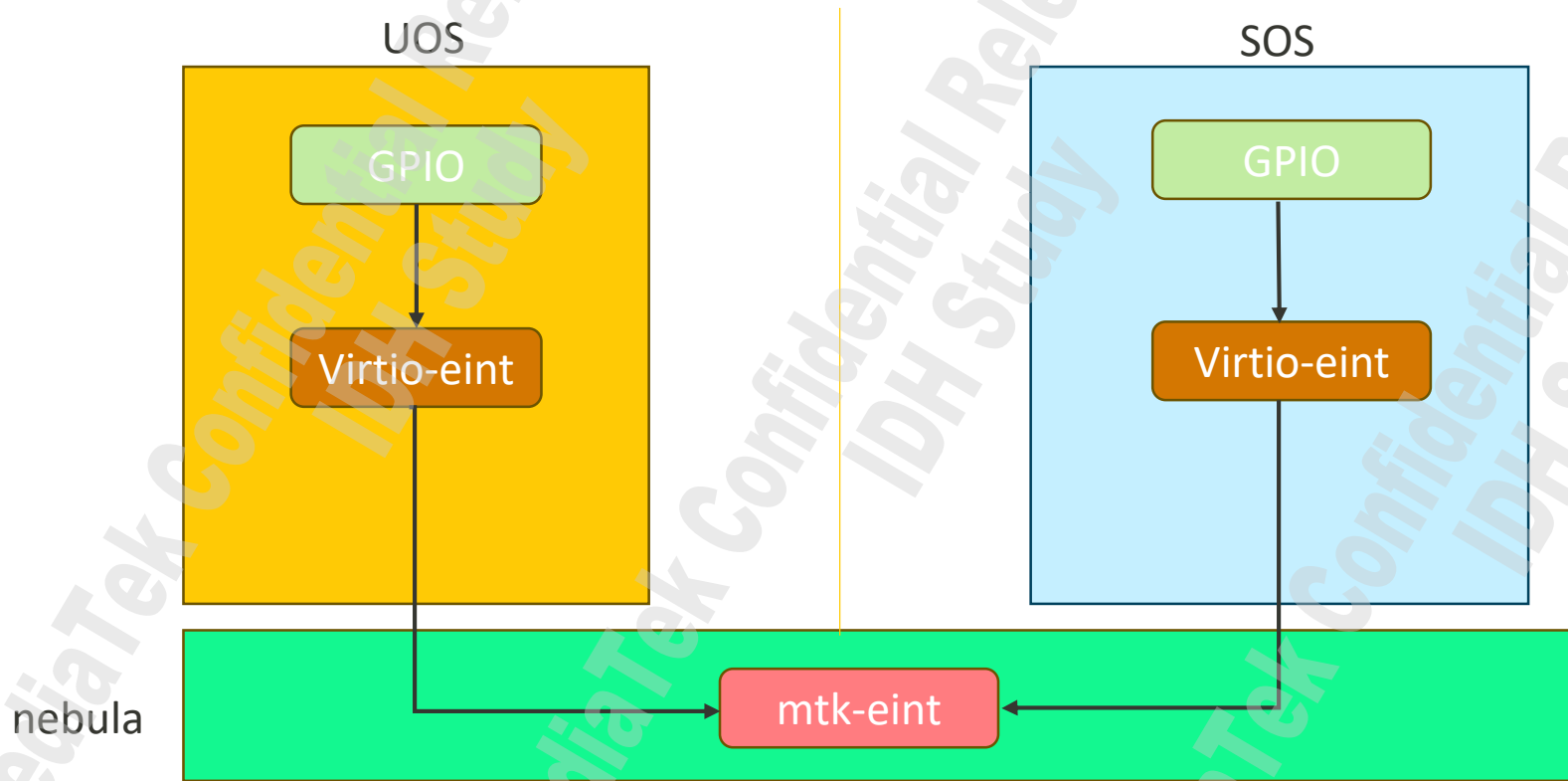


Virtqueue

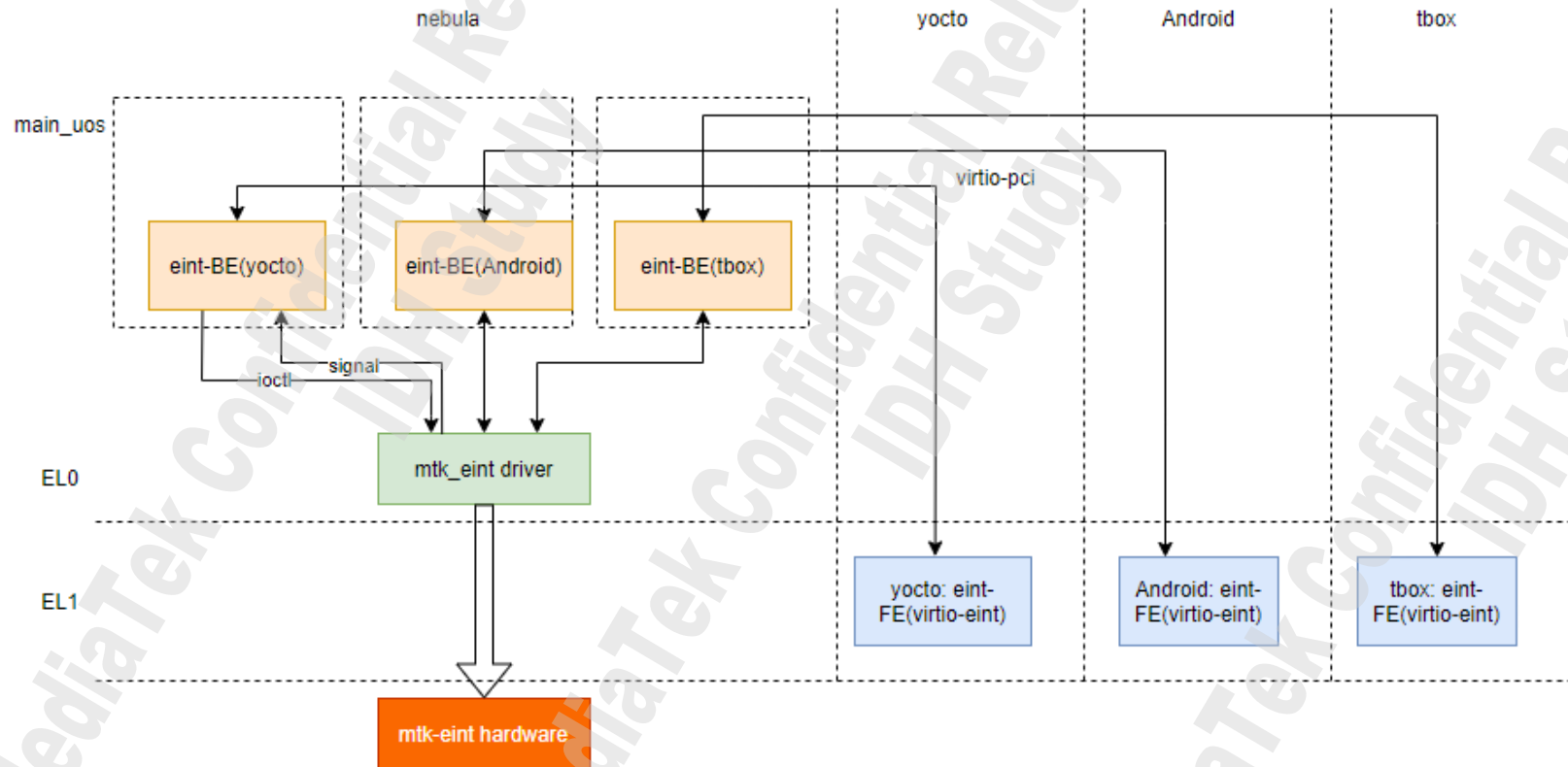




Virtual Eint(sink nebula)



Nebula-sink eint



Eint Register Introduction

```
static const struct mtk_eint_regs mtk_generic_eint_regs = {  
    .stat      = 0x000,  
    .ack       = 0x040,  
    .mask      = 0x080,  
    .mask_set  = 0x0c0,  
    .mask_clr  = 0x100,  
    .sens      = 0x140,  
    .sens_set  = 0x180,  
    .sens_clr  = 0x1c0,  
    .soft      = 0x200,  
    .soft_set  = 0x240,  
    .soft_clr  = 0x280,  
    .pol       = 0x300,  
    .pol_set   = 0x340,  
    .pol_clr   = 0x380,  
    .dom_en    = 0x400,  
    .dbnc_ctrl = 0x500,  
    .dbnc_set  = 0x600,  
    .dbnc_clr  = 0x700,  
    .raw_stat  = 0xa00,  
};
```

DebugTips: confirm eint irq count

```

CPU0      CPU1      CPU4      CPU5      CPU6
root@auto8678p164sos:/# cat /proc/interrupts |grep eint
47:         0         0         0         0         0 mtk-eint    1 Edge      16310000.mmc_cd
50:         0         0         0         0         0 mtk-eint    4 Edge      ilitek_touch_irq
58:         0         0         0         0         0 mtk-eint   12 Level    ap_eint
78:         1         0         0         0         0 mtk-eint   32 Edge      gt9xx_DSI-2
106:        0         0         0         0         0 mtk-eint   60 Edge      Volume_Down
120:        0         0         0         0         0 mtk-eint   74 Edge      gt9xx_DP-1
163:        1         0         0         0         0 mtk-eint  117 Level    gEint
291:        2         0         0         0         0 mtk-eint  245 Edge      MT6376_FAULT
297:        0         0         0         0         0 mtk-eint  251 Edge      gEint
299:        0         0         0         0         0 mtk-eint  253 Edge      gEint
300:        0         0         0         0         0 mtk-eint  254 Edge      odm:extcon-usb2
302:        2         0         0         0         0 mtk-eint  256 Edge      odm:extcon-usb2
304:        0         0         0         0         0 mtk-eint  258 Edge      gt9xx_eDP-1
337:        1         0         0         0         0 mtk-eint  291 Level    rcs_irq
338:        0         0         0         0         0 mtk-eint  292 Level    rcs_irq

```

Proc.c: show_interrupts parse irq_desc

DebugTips: dump register

```
echo <eint_num> > /sys/bus/platform/drivers/mt6991-pinctrl/eint_pin_status
```

```
cat /sys/bus/platform/drivers/mt6991-pinctrl/eint_pin_status
```

```
rl/eint_pin_statusos:/# echo <eint_num> > /sys/bus/platform/drivers/mt6991-pinctrl/eint_pin_status
n_statuso8678p164sos:/# echo 32 /sys/bus/platform/drivers/mt6991-pinctrl/eint_pin_status
32 /sys/bus/platform/drivers/mt6991-pinctrl/eint_pin_status
atus@auto8678p164sos:/# cat /sys/bus/platform/drivers/mt6991-pinctrl/eint_pin_status
=====EINT Pin Dump=====
Pin=0(enabled)
Type=level_low
Pending=0
Raw=0
Soft=0
Mask=1
```

DebugTips: confirm whether it is a virtio EINT

3os devices:

```
D:\>adb -s 0123456789ABCDEF shell
auto8678p1_64_bsp_vm_uos_b0:/ $ su
auto8678p1_64_bsp_vm_uos_b0:/ # cat /pr
proc/
auto8678p1_64_bsp_vm_uos_b0:/ # cat /proc/interrupts |grep eint
141:      0          0          0          0          0 mtk-eint   60 Edge      Volume_Down
326:      0          0          0          0          0 mtk-eint  245 Edge      MT6376_FAULT
332:      0          0          0          0          0 mtk-eint  251 Edge      odm:extcon-usb1
334:      0          0          0          0          0 mtk-eint  253 Edge      odm:extcon-usb1
auto8678p1_64_bsp_vm_uos_b0:/ # exit
auto8678p1_64_bsp_vm_uos_b0:/ $ exit

D:\>adb -s YOCTO shell
root@auto8678p164s0sb0:/# cat /proc/interrupts |grep eint
47:      0          0          0          0          0 mtk-eint   1 Edge      16310900.mmc-ed
50:      0          0          0          0          0 mtk-eint   4 Edge      ilitek_touch_irq
58:      0          0          0          0          0 mtk-eint  12 Level     ap_eint
78:      0          0          0          0          0 mtk-eint  32 Edge      gt9xx_DSI-2
106:     0          0          0          0          0 mtk-eint  60 Edge      Volume_Down
120:     0          0          0          0          0 mtk-eint  74 Edge      gt9xx_DP-1
163:     8          0          0          0          0 mtk-eint 117 Level     gEint
291:     2          0          0          0          0 mtk-eint 245 Edge      MT6376_FAULT
297:     0          0          0          0          0 mtk-eint 251 Edge      gEint
299:     0          0          0          0          0 mtk-eint 253 Edge      gEint
300:     0          0          0          0          0 mtk-eint 254 Edge      odm:extcon-usb2
302:     0          0          0          0          0 mtk-eint 256 Edge      odm:extcon-usb2
304:     0          0          0          0          0 mtk-eint 258 Edge      gt9xx_eDP-1
337:     0          0          0          0          0 mtk-eint 291 Level     rcs_irq
338:     0          0          0          0          0 mtk-eint 292 Level     rcs_irq
root@auto8678p164s0sb0:/# exit
exit

D:\>adb -s YOCTO.TBOX shell
root@auto8678p164u0stbox:/# cat /proc/interrupts |grep eint
118:     0          0          0          0          0 mtk-eint   60 Edge      Volume_Down
176:     0          0          0          0          0 mtk-eint 117 Level     17-001b
303:     0          0          0          0          0 mtk-eint 245 Edge      MT6376_FAULT
root@auto8678p164u0stbox:/#
```

DebugTips: no interrupt

- 1: 确认GPIO DIN值是否改变
- 2: dump 确认eint 对应gpio status bit
- 3: 是否edge设置debounce
- 4: 确认domain reg是否有设置
- 5: 请GIC协助确认是否有收到
- 6: virt-eint 确认virtqueue异常或者worker没有执行

DebugTips: interrupt storm

1: 设置type和gpio默认状态一致

2: level sense, gpio状态没有复位

3: ...

Code path

name	Os	Code path	CONFIG
Deint	android	alps/vendor/mediatek/proprietary/trustzone/tf-a-2.xx/plat/mediatek/drivers/deint	N/A
cirq	android	alps/vendor/mediatek/proprietary/trustzone/tf-a-2.xx/plat/mediatek/drivers/cirq	CPU_PM_IRQ_REMAIN_ENABLE
Eint event	android	alps/vendor/mediatek/proprietary/bootable/bootloader/preloader/platform/mtxxxx/src/drivers/platform.c unmask_eint_event_mask	N/A
scp eint	android	alps/vendor/mediatek/proprietary/tinysys/common/drivers/eint/v02/src	CFG_EINT_SUPPORT
Virtio-eint	Android yocto	alps/kernel/kernel_device_modules-6.xx/drivers/pinctrl/mediatek src/kernel/linux/v6.xx_mtxxxx/co_device_module/drivers/pinctrl/mediatek	CONFIG_VIRTIO_EINT_MTK
vhosteint	yocto	src/kernel/modules/(mtxxxx/virt/grt or virt_mtk)/eint	N/A
Mtk-eint	Android yocto	alps/kernel/kernel_device_modules-6.xx/drivers/pinctrl/mediatek src/kernel/linux/v6.xx_mtxxxx/co_device_module/drivers/pinctrl/mediatek	N/A
Eint vmm in sdk	yocto	src/hypervisor/grt/thyp-sdk/vmm/nbl_vmm/devices/eint.cc	vendor/etc/hyper_android/uos_alps_pv8668.lua uos_config:setFlag("eint", true)
Eint vmm in ddk	yocto	src/hypervisor/grt_ddk/garnet/drivers/board/mtxxxx/mtxxx-eint-hw.c	N/A

Q&A

- N/A

The End